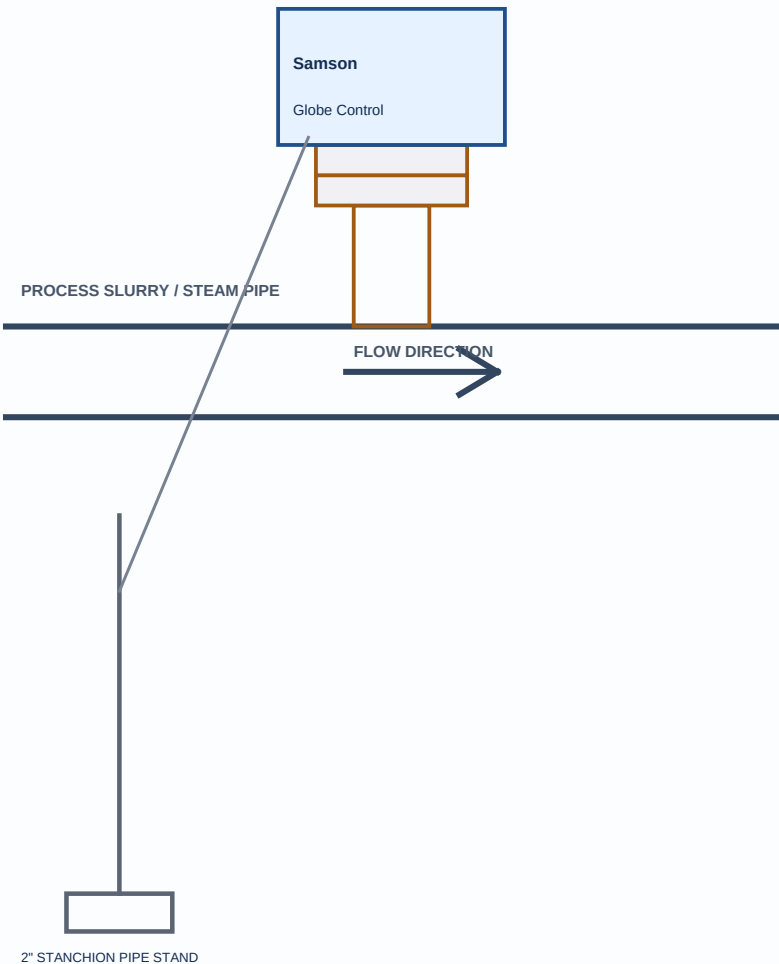


TEMPLATE TYP-21: GLOBE CONTROL VALVE

TYPICAL TAGS: TCV-40222 / TV-60202

SERVICE: High-Precision Steam & Slurry Flow Modulation | PLC I/O: 1756-OF8 (AO Setpoint) & 1756-IF16 (AI Feedback)

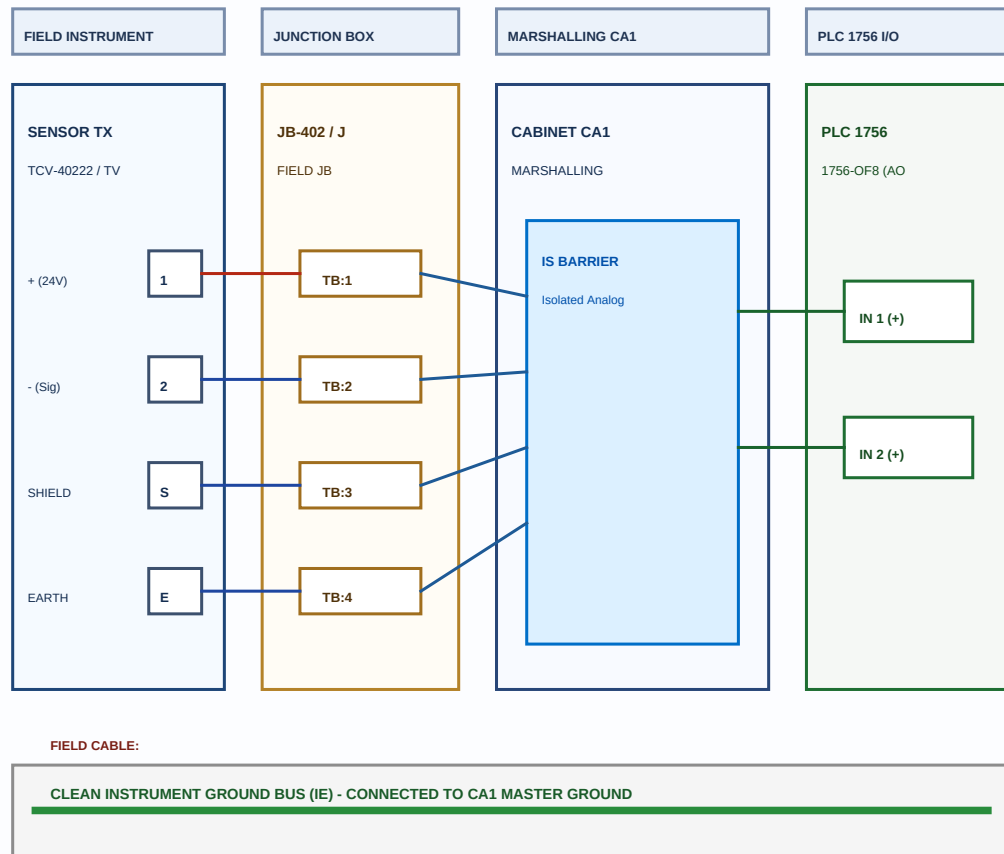
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Flange ASME B16.5 Cl.150/300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP66 Die-cast Aluminum Posit..
Hazardous Class:	ATEX II 2G Ex ia IIC T6 / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AO: 4-20mA Valve Cmd (0-100%) / AI: Po..
Power Supply:	2-Wire 4-20mA Loop + 6.0 bar Instrumen..
Field Cable Type:	2Px1.5mm ² Twisted Shielded (Belden 8777)
Destination JB:	JB-402 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Isolated Analog Output / IS Repeater
PLC Module:	1756-OF8 (AO Setpoint) & 1756-IF16 (AI..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SAMSON

MODEL: Type 3241 Globe Valve + Trovis 3730-1 Smart Positioner

SERVICE: High-Precision Steam & Slurry Flow Modulation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Globe Control Valve Type 3241 Samson Cast Steel / SS316 Body		1 EA
2	Pneumatic Diaphragm Actuator T Samson Multi-Spring Actuator		1 EA
3	Smart Positioner Trovis 3730-1 HART Communication, Auto-Tuning		1 EA
4	SMC Air Filter Regulator AW20 Clean Air 1/4" w/ Pressure Gauge		1 SET
5	Cable Glands M20x1.5 Ex d/e IP Nickel-plated Brass Glands		2 EA

ENGINEERING INSTALLATION NOTES

- Perform Trovis 3730-1 automatic initialization routine after mechanical mounting is completed.
- Connect 4-20mA setpoint from PLC 1756-OF8 to Positioner Terminals 11(+) and 12(-).
- Ensure air supply pressure to positioner is regulated strictly between 4.0 and 6.0 bar.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTMODULATING GLOBE CONTROL VALVE HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-021
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
23 / 25